

1 Problem setup

We are solving the following dimensionless system:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \sqrt{\frac{Pr}{Ra}} \nabla^2 \mathbf{u} + \theta \mathbf{e}_y \quad \text{in } \Omega \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega \quad (2)$$

$$\frac{\partial \theta}{\partial t} + \mathbf{u} \cdot \nabla \theta = \frac{1}{\sqrt{Ra Pr}} \nabla^2 \theta \quad \text{in } \Omega \quad (3)$$

$$\mathbf{u} = 0 \quad \text{on } \Gamma \quad (4)$$

$$\theta = \frac{1}{2} \quad \text{on } \Gamma_{\text{left}} \quad (5)$$

$$\theta = -\frac{1}{2} \quad \text{on } \Gamma_{\text{right}} \quad (6)$$

$$\frac{\partial \theta}{\partial y} = 0 \quad \text{on } \Gamma_{\text{top, bottom}} \quad (7)$$

Where (1) is the incompressible Navier-Stokes equation, (2) is the continuity equation, and (3) is the energy equation written in terms of temperature θ . We are going to solve this system using finite elements, so we need the weak formulation of the equations. We obtain these by multiplying by test functions from suitable test spaces and integrating over Ω .

$$\int_{\Omega} \frac{\partial \mathbf{u}}{\partial t} \cdot \mathbf{v} + (\mathbf{u} \cdot \nabla \mathbf{u}) \cdot \mathbf{v} - p \nabla \cdot \mathbf{v} + \sqrt{\frac{Pr}{Ra}} \nabla \mathbf{u} : \nabla \mathbf{v} - \theta \mathbf{v}_y \, dV = 0 \quad (8)$$

$$\int_{\Omega} (\nabla \cdot \mathbf{u}) \phi \, dV = 0 \quad (9)$$

$$\int_{\Omega} \frac{\partial \theta}{\partial t} \psi + (\mathbf{u} \cdot \nabla \theta) \psi + \frac{1}{\sqrt{Ra Pr}} \nabla \theta \cdot \nabla \psi \, dV = 0 \quad (10)$$

These equations should hold for all $(\mathbf{v}, \phi, \psi) \in W_0^{1,2}(\Omega) \times L^2(\Omega) \times W_0^{1,2}(\Omega)$. This gives us the foundation we need to solve this problem in Firedrake. Per the problem specifications, the domain is a rectangle with a height to width ratio of 8:1. We included five specific points in the mesh used for measurements, namely the point $a = (0.181, 7.37)$. The geometry was generated using Netgen and imported into Firedrake. The characteristic mesh size was given by $h = \frac{1}{10n}$. The final simulation ran with $n = 10$.

The finite element spaces used for the simulations were continuous Galerkin of second order for velocity and temperature and first order for the pressure.

The initial conditions are:

$$\mathbf{u}(\mathbf{x}, 0) = 0 \quad (11)$$

$$\theta(\mathbf{x}, 0) = 0 \quad (12)$$

We measured several quantities to compare with existing results at a point a , namely the average temperature, average velocity in the x-direction, and the Nusselt number along the heated/cooled wall. The Nusselt number was calculated using the relation:

$$Nu(t) = \frac{1}{H} \int_0^H \left| \frac{\partial \theta}{\partial x} \right|_{x=0,W} dx$$

We ran the simulation until $t = 180$ with a time step of $dt = 0.1$. The measurements were taken starting from $t = 100$ to avoid the initial turbulent phase.

The measurements we obtained from this simulation were in good agreement with the benchmark results, as can be seen in the following table:

	Value	Reference value	Relative error [%]
Temperature	0.2633	0.2647	0.54
Velocity	0.0545	0.0572	4.78
Nusselt number	4.5796	4.5791	0.01

We can see the oscillations as well as the overall tendency for the average to move towards the goal value in the following graphs of the quantities of interest.



